

Cyclistic Case Study – Executive Summary

Business Problem

Cyclistic, a bike-share company, wants to **convert casual riders into annual members** to increase revenue and long-term customer loyalty.

Data & Preparation

- **Dataset:** 12 months of trip data (5.5M+ rides).
- **Cleaning:** Removed duplicates, standardized formats, calculated ride length, filtered outliers (>24 hrs).
- **Transformation:** Added derived fields (ride length in minutes, day of week, month).

Key Analysis Findings

- **Ride Length:** Casual riders average **20–25 min**, members **12–14 min**.
- **Day of Week:** Members ride more on **weekdays** (commuting), casual riders peak on **weekends** (leisure).
- **Seasonal Trends:** Casual rider activity spikes in **summer months**, members remain steady year-round.
- **Station Usage:** Casual riders favor **tourist areas/parks**, members prefer **business districts/residential zones**.

Visual Storytelling

- Bar chart: Average ride length by rider type.
- Line chart: Ride counts by weekday vs weekend.
- Line chart: Monthly ride counts (seasonal trends).
- Bar chart: Top start stations by rider type.

Final Recommendations

1. **Target casual riders with membership incentives** (summer promotions, weekend discounts).
2. **Highlight commuter benefits** (cost savings, convenience, employer partnerships).
3. **Leisure campaigns** around parks/tourist areas to capture weekend riders.
4. **Digital outreach** via app notifications, email, and social media.

Portfolio Value

This case study demonstrates:

- **Technical skills:** Data wrangling, cleaning, visualization (Python, Pandas, Seaborn).
- **Analytical skills:** Identifying behavioral differences between rider types.
- **Business impact:** Actionable recommendations aligned with Cyclistic's strategic goals.

- **Communication skills:** Clear documentation, executive-style storytelling, and concise insights.